

includes noise reduction, dynamic backlight control, contrast enhancements... the whole lot. Since you'll be performing this calibration by hooking up your display to a PC/laptop, make sure you set the TV up to disable image scaling and enable 1:1 pixel mapping. Sadly, enabling this setting is not that easy because some TV's flat out lack this crucial feature, whereas others have it hidden away in the deeper sub-menus under an obscure name. Some manufacturers dub 1:1 pixel mapping as "Aspect Ratio: Normal", whereas others call it "PC Mode", "Screen Fit", or "Dot by Dot". If you can't find this setting, most displays come with detailed manuals (which are sometimes online) that explain how to enable it. Note: this setting doesn't apply to PC monitors.

Thereafter, change the HDMI colour mode from analogue YCbCr to digital RGB mode. Also make it a point to go through the GPU's driver setting to disable all colour and image enhancements. Finally, make sure you set the colour temperature or white point to 6500 K, which should be called "normal" or "mid" setting depending upon your TV's make.

Now that everything's primed and ready, make sure that you have left your display switched on for at least 30 minutes before loading up <http://www.lagom.nl/lcd-test/> for the final calibration. What follows is a detailed explanation on what the important test patterns entail and how you can go about adjusting your display settings optimally.

◆ **Black Level**

This evaluates the capability of the television to display deep blacks. The image is comprised of multiple swatches of progressively black tints that finally blend into the perfect black background. The idea is to first coarsely adjust the brightness setting until the background stops being grey and appears black. You know you have got the brightness setting right when any further reduction in the level doesn't make any difference to the blackness of the background. Once the brightness levels are optimally set up, look out for the first ten tints of RGB value 1 to 10. If your display has an excellent panel, it should be able to differentiate all ten swatches from the black background. If not, increase the brightness until you can tell the darker swatches from the background. Make sure that you aren't turning the black background grey by increasing the brightness too much.

◆ **White Saturation**

Like the last test image, this one allows the ability of the display to distinguish between extreme white details. This is achieved by setting